

Practical Analysis of WPA2 and WPA3 Resistance to Dictionary-Based Brute-Force Attacks Using Weak Passwords

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Abstract—Wireless network security is fundamentally based on encryption and authentication mechanisms that prevent unauthorized access to data and network resources. Among these, the Wi-Fi Protected Access (WPA) family of protocols, specifically WPA2 and its successor WPA3, are widely adopted standards designed to secure wireless communications through robust encryption and key-exchange methods. This study presents a practical evaluation of WPA2 and WPA3 against dictionary-based brute-force attacks using weak or common passwords. The experimental phase involved configuring two wireless networks under WPA2-Personal and WPA3-Personal security standards. For WPA2, Aircrack-ng and Hashcat were used to capture the four-way handshake and perform offline dictionary attacks. For WPA3, due to its resistance to offline attacks through the SAE protocol (Simultaneous Authentication of Equals), real-time attacks were conducted using Wacker. All experiments were carried out in a controlled Rocky Linux environment, recording key metrics such as cracking time and attack success rate. The results show that while WPA3 provides greater resistance to offline attacks, it remains vulnerable when weak passwords are used. These findings highlight the importance of proper credential management, even when modern security protocols are in place.

Keywords—Wi-Fi Protected Access II, Wi-Fi Protected Access III, Wireless Network Security, Dictionary-Based Attack, Brute-Force Password Attack

I. INTRODUCTION

Wireless communication is an essential part of modern digital infrastructure, supporting everything from home networks to enterprise and industrial systems. As the reliance on wireless networks increases, so does the importance of ensuring their security, particularly in the authentication mechanisms that protect user access. Wi-Fi Protected Access (WPA) protocols, specifically WPA2 and its successor WPA3, have been designed to provide strong encryption and authentication in these environments [1].

Wireless networks, due to their open and easily accessible nature, are especially vulnerable to remote attacks, as attackers

do not need to be physically within the coverage area to intercept traffic. Information circulating on the network can be detected, captured, and analyzed without alerting users, making security a primary concern in Wi-Fi environments beyond performance or coverage. Despite this, many users are unaware of these risks and often underestimate the importance of properly protecting their networks [2].

WPA2, introduced in 2004, relies on the four-way handshake mechanism and the pre-shared key (PSK) approach, making it vulnerable to dictionary-based brute-force attacks when users select weak passwords [3] [4]. WPA3, released in 2018, sought to address these vulnerabilities by implementing the Simultaneous Authentication of Equals (SAE) protocol, which prevents offline attacks by requiring a unique key exchange for each authentication attempt [5].

Although WPA3 has incorporated this security enhancement, a fundamental vulnerability remains: the use of weak passwords. The OWASP Top 10, a widely recognized standard on web application security risks, includes “A07:2021 – Identification and Authentication Failures” [6] as one of the most critical issues. This category highlights the risks associated with poor credential management, including the use of weak or easily predictable passwords. Although OWASP is primarily focused on web applications, the principles it establishes are equally applicable to wireless networks, where authentication mechanisms rely heavily on user-defined passwords. Regardless of the protocol’s technical advancements, an inadequate password policy still represents a considerable risk in both web and network environments.

This article presents a practical analysis comparing the resistance of WPA2 and WPA3 to dictionary-based brute-force attacks with weak passwords. Test networks were set up with both security features, attacked by Rocky Linux machines, obtaining results such as cracking time and successful attempts.

The results of this research aim to contribute to the understanding of how the selection of weak passwords weakens even

the most advanced authentication protocols. This underscores that user practices remain a critical component of wireless network security.

II. BACKGROUND

A. Overview of WPA2 and WPA3 Security Protocols

WPA2 is a descended security protocol from WPA, both of which have been developed by the Wi-Fi Alliance to protect wireless networks. This protocol contains improvements such as the AES (Advanced Encryption Standard) encryption protocol, making it more secure than the TKIP (Temporal Key Integrity Protocol) used in WPA [7]. It is vulnerable to dictionary attacks due to pre-shared keys (PSK), as well as deauthentication attacks.

Today, the likelihood of wireless devices being compromised has increased significantly due to multiple weaknesses in WLAN networks. Despite the existence of encryption protocols such as WPA or WPA2, many users are unaware of adequate protection mechanisms, leaving their networks vulnerable to attacks [8].

WPA3, introduced in 2018, is an improvement over WPA2 with respect to the use of Simultaneous Authentication of Equals (SAE) handshake, making online dictionary attacks ineffective and improving the privacy of transmitted data. [9]

B. Dictionary and Brute-Force Attacks in Wireless Networks

Dictionary attacks are a hacking method that is used to crack passwords or passphrases. They involve trying a predefined list of words or character combinations [10]. According to a study analyzing more than 600 million leaked passwords, only a fraction of the words in dictionaries such as the Mac system dictionary or the leaked Rockyou archive were actually used by users. Specifically, approximately 90,000 of the 235,886 words in the Mac dictionary and about 6 million of the 13 million unique Rockyou passwords discovered in leaked passwords [11].

Also, a brute-force attack is when an attacker attempts to access a network or system by trying to find all passwords. It is an effective attack, but terribly slow and requires significant resources if the password is complex [10].

C. Authentication Weaknesses and OWASP Principles

The OWASP TOP 10 is a classification that identifies the main security risks in web applications. It was developed by the Open Worldwide Application Security Project (OWASP) and contains 10 vulnerability categories, including details of risks, examples of real-world attacks, and recommended practices for mitigation.

Although this classification is geared toward the web application sector, the "A07:2021 - Identification and Authentication Failures" [12] vulnerability emphasizes the problems that weak passwords and a lack of control over brute-force attacks can cause.

Weak passwords protected by protocols such as WPA2 or WPA3 align with OWASP category A07. Even with advanced security mechanisms, if users choose easy-to-predict passwords, networks can be easily compromised.

D. Related Works

To begin, we have the article "Evolution of Wi-Fi Protected Access: Security Challenges" [13], which explores the evolution of Wi-Fi security protocols. Researchers emphasize the improvements that WPA3 has brought over WPA2, where WPA3 hides the traffic sent data, in addition to implementing the Dragonfly protocol, which requires the exchange of random numbers before calculating the secret key, thus preventing dictionary attacks. The article concludes by highlighting the advantages of the WPA3 standard over WPA2 but acknowledges the significant challenge regarding user adoption.

Also, focusing more on the topic of brute-force attacks, we have the article "Wi-Fi Handshake analysis of password patterns in Wi-Fi networks [14]", which presents an analysis of password patterns in Wi-Fi networks. Using HASHCAT software, an audit was guided to capture Wi-Fi network handshakes and crack passwords using a dictionary developed from contextual data related to the location where the analysis was conducted. The results were a 33% password recovery success rate, highlighting a vulnerability in common passwords.

Finally, the article "Penetration Testing and Machine Learning-Driven Cybersecurity Framework for IoT and Smart City Wireless Networks" [15] proposes a smart city environment integrating an IOT infrastructure, focusing on the cybersecurity of these systems by addressing the vulnerabilities that this environment would present. It uses a framework that combines penetration testing, anomaly detection, and predictive modeling. The results regarding the penetration tests on the network where WPA2-Personal is used were the capture and decryption of the WPA2 handshake, demonstrating that a network with this security protocol and a weak password was very susceptible to dictionary attacks, recommending the implementation of more robust passwords and the use of a more robust protocol, such as WPA3.

III. METHODOLOGY

A. Type of Study

This research adopts a quantitative, applied, and experimental approach, aimed at practically evaluating the resistance of the WPA2 and WPA3 protocols to dictionary-based brute-force attacks. A controlled laboratory environment was designed to simulate real-world wireless networks under repeatable conditions, to empirically observe the protocols' behavior against weak passwords.

B. Conceptual Definition and Organization of Variables

The variables considered in the study were organized according to their relationship with the security of the protocol against authentication attacks:

C. Methods, Techniques and Instruments

The attack simulations were conducted on a machine running the Rocky Linux operating system and two wireless networks, with WPA2-Personal and WPA3-Personal security respectively, using three weak passwords: one with numbers, one with letters, and one with numbers and letters.

TABLE I
DEFINITION OF STUDY VARIABLES

Variable	Conceptual Def.	Operational Def.
Cracking time	Duration of the attack process until the password is obtained	Time recorded from the start of the attack until successful password recovery
Attack Success Rate	Effectiveness of the attack against weak passwords	Percentage of compromised passwords relative to the total number of attempts
Security Protocol	Evaluated encryption standard	WPA2-Personal, WPA3-Personal

The tool used to attack the WPA2 network was Aircrack-ng, to capture the WPA2 handshake and perform offline attacks.

For WPA3, since offline attacks are not possible, the Wacker tool was used, which performs real-time attacks during the SAE (Simultaneous Authentication of Equals) authentication process. The techniques applied were:

- Packet capture and authentication process in WPA2 networks using monitor mode and controlled deauthentication.
- Execution of offline dictionary attacks with Aircrack-ng on captured handshakes.
- Simulation of WPA3 authentication attempts using Wacker, evaluating resistance to weak passwords.
- Logging and analyzing time and success metrics.

D. Experimental procedure

1) *Network Configuration*: Two independent wireless networks were created, one configured with WPA2-Personal and the other with WPA3-Personal, both protected with known, weak passwords.

2) *Authentication Process Capture or Execution*: In WPA2, aircrack-ng was used to capture the handshake. In WPA3, Wacker was used to attempt real-time authentication.

3) *Dictionary Attack*: In WPA2, an offline attack was executed using the capture file and rockyou.txt dictionary. In WPA3, multiple connections were attempted using Wacker with entries from the same dictionary.

4) *Result Log*: The time required for decryption, the number of successful attempts, failures, and the protocol's behavior in each scenario were documented.

IV. RESULTS AND DISCUSSIONS

A. Overview of Experimental Results

The experimental evaluation compared the resistance of WPA2 and WPA3 security protocols against dictionary-based brute-force attacks using weak passwords. Three commonly used passwords — "12345678", "iloveyou", and "abcd1234" — were tested across 10 attack iterations in each protocol. The

key metric recorded was time to crack (in seconds), allowing a comparative analysis of attack feasibility and response behavior between both protocols.

B. WPA2 Results

In the case of WPA2, all three passwords were successfully cracked in negligible time using offline attacks with Aircrack-ng. For "12345678" and "iloveyou", the cracking time was consistently below 0.1 seconds across all attempts, confirming the protocol's vulnerability when weak passwords are used. The password "abcd1234" showed one outlier with a cracking time of 45 seconds, which may indicate slight variations in dictionary position or processing.

TABLE II
WPA2 CRACKING TIMES (IN SECONDS)

Password	"12345678"	"iloveyou"	"abcd1234"
Time 1	0.0 s	0.0 s	0.0 s
Time 2	0.0 s	0.0 s	0.0 s
Time 3	0.0 s	0.0 s	0.0 s
Time 4	0.0 s	0.0 s	0.0 s
Time 5	0.0 s	0.0 s	0.0 s
Time 6	0.0 s	0.0 s	0.0 s
Time 7	0.1 s	0.0 s	0.0 s
Time 8	0.0 s	0.0 s	0.0 s
Time 9	0.0 s	0.0 s	45.0 s
Time 10	0.0 s	0.1 s	0.0 s

C. WPA3 Results

For WPA3, real-time attacks using Wacker showed significantly higher resistance. The cracking times for "12345678" and "iloveyou" ranged from 1 to 9 seconds, with minor variance. However, the password "abcd1234" exhibited greater inconsistency, with cracking times fluctuating between 1.25 seconds and 43 seconds. These variations reflect the protocol's Simultaneous Authentication of Equals (SAE) mechanism, which requires a new handshake attempt for each password guess, increasing computational cost and slowing brute-force feasibility.

TABLE III
WPA3 CRACKING TIMES (IN SECONDS)

Password	"12345678"	"iloveyou"	"abcd1234"
Time 1	2.0 s	1.0 s	25.0 s
Time 2	9.0 s	1.0 s	1,520 s
Time 3	5.0 s	1.0 s	1,250 s
Time 4	1.0 s	1.0 s	1,270 s
Time 5	1.0 s	1.0 s	1,290 s
Time 6	1.0 s	2.0 s	43.0 s
Time 7	1.0 s	1.0 s	1,250 s
Time 8	1.0 s	1.0 s	1,270 s
Time 9	1.0 s	1.0 s	1,420 s
Time 10	1.0 s	1.0 s	39.0 s

D. Comparative Analysis

The cracking success rate was 100% for both WPA2 and WPA3 across all passwords tested. This result demonstrates that despite security improvements introduced in WPA3, both protocols remain highly vulnerable when users use weak or

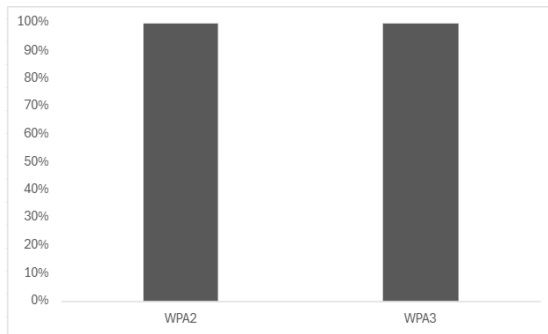


Fig. 1. Cracking Success Rate.

commonly used passwords. The fact that all passwords were ultimately compromised, even for WPA3 with its SAE-based authentication, demonstrates that dictionary-based brute-force attacks are not only feasible but also highly effective in practice.

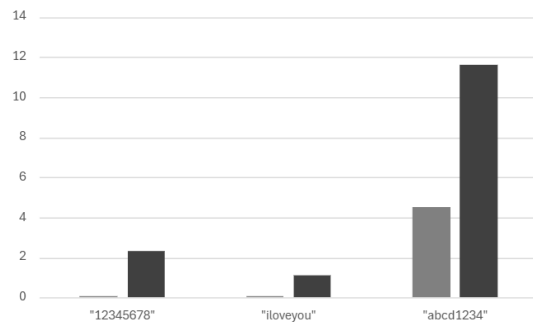


Fig. 2. WPA2 vs WPA3 comparison.

Average cracking times for WPA3 are seconds longer than those for WPA2 for all passwords analyzed. In fact, the bars representing WPA2 are almost imperceptible in the graph due to their extremely low values, especially those below one second. This sharpness contrasts WPA3's greater resistance to brute-force dictionary attacks. The password "abcd1234," in particular, shows a clear separation between the protocols, with WPA3 requiring more than twice the average time than WPA2. Despite this, both protocols succumbed to the attacks, underscoring that WPA3's improvements slow the attack but do not completely prevent it when weak passwords are used.

E. Discussions

The experimental results confirm that WPA2 remains vulnerable to dictionary-based brute-force attacks due to its reliance on offline four-way handshake capture and PSK authentication. This aligns with the findings of Asante and Akomea-Agyin, who emphasized that WPA2-PSK's architecture allows attackers to exploit weak passwords effectively through offline means [3]. The extremely short cracking times observed for "12345678" and "iloveyou" reinforce earlier conclusions by Carballal et al., who reported high success rates when exploiting predictable password patterns in WPA2 networks

using HASHCAT and rockyou-based dictionaries [14]. These consistent vulnerabilities point to the enduring risks posed by poor user practices, even under technically robust encryption schemes.

Although WPA3 showed greater resistance due to the Simultaneous Authentication of Equals (SAE) protocol, which prevents offline attack vectors, it was still compromised when weak passwords were used. This is consistent with the broader concern expressed in the OWASP Top 10 under the A07:2021 category, which identifies authentication failures and weak credential management as critical risk factors [12]. Moreover, Zhukabayeva et al. demonstrated through penetration testing that even in advanced IoT infrastructures, WPA2 networks were compromised due to poor password hygiene, and they recommended transitioning to WPA3 alongside stronger passwords [15]. Our results extend this concern, showing that even WPA3's improvements cannot compensate for weak password use, highlighting that user behavior remains a dominant factor in security outcomes.

A more detailed analysis of the results reveals important differences between the protocols. In WPA2, the three tested passwords ("12345678", "iloveyou", and "abcd1234") were consistently cracked almost instantaneously, with average cracking times below 0.1 seconds, regardless of complexity. By contrast, WPA3 exhibited significant variability: simple numeric or dictionary-based words were compromised in only 1–9 seconds, while the alphanumeric password "abcd1234" produced times ranging from 1.25 seconds to over 1500 seconds across different attempts. This variability highlights the fact that WPA3 penalizes weak password complexity more strongly than WPA2, yet still fails to provide real protection if the chosen password remains predictable or common.

From a comparative standpoint, both protocols ultimately reached a 100% cracking success rate in all scenarios, but WPA3 introduced a computational cost that could deter low-resource attackers. However, in practical terms, determined adversaries with sufficient time or resources can still achieve success. This reinforces the conclusion that protocol design alone cannot ensure security, and robust password policies are indispensable.

It is also important to acknowledge the limitations of this study. The experiments were performed in a controlled laboratory environment, without external interference or realistic network congestion, which may not fully replicate operational conditions. Furthermore, only three weak passwords were evaluated, providing a representative but narrow perspective. The dictionary used (rockyou.txt) is widely employed in password cracking studies, but real attackers may employ larger, hybrid, or customized dictionaries. Finally, no evaluation of hardware acceleration (e.g., GPU clusters, FPGA-based cracking) was performed, which could drastically reduce attack times in practice. These limitations suggest that while the study provides valuable insights into protocol weaknesses, further research is needed to explore a broader range of conditions, passwords, and adversary capabilities.

In summary, the discussion underscores two key points:

(1) WPA2 remains critically insecure when weak passwords are used, being compromised almost instantly, and (2) WPA3 introduces measurable resistance but cannot mitigate the fundamental problem of poor credential management. These findings align with prior literature and OWASP recommendations, highlighting the necessity of strong password policies and complementary authentication practices beyond protocol adoption.

V. CONCLUSIONS

The experimental results show that the attack success rate was 100% for both WPA2 and WPA3 when weak passwords were used. This demonstrates that, despite improvements in WPA3's security architecture, the use of common or predictable passwords remains a critical vulnerability. Poor credential management can compromise the security of a protocol, which aligns with the OWASP Top 10 risk A07:2021, highlighting authentication failures related to weak passwords and exposed credentials.

A comparison of average cracking times shows a significant difference between the two protocols: WPA2 was breached almost instantaneously, while WPA3 showed some resistance, especially against more complex combinations such as "abcd1234." However, the difference in time does not provide effective protection if passwords remain weak.

These findings reinforce the importance of not relying solely on the implemented security protocol. Achieving true security in wireless networks requires strong password policies, additional protection mechanisms, and proper authentication management aligned with best practices, ensuring a comprehensive approach against persistent threats.

The study consistently demonstrates that the security of wireless networks, even under advanced protocols like WPA3, is vulnerable to dictionary-based brute-force attacks when weak or commonly used passwords are employed. To mitigate these risks, it is essential to adopt strong password policies that prohibit predictable passwords, require sufficient length and complexity, and encourage user awareness regarding credential management.

Additionally, integrating additional protection mechanisms such as multi-factor authentication provides a complementary layer of security, reducing the risk of unauthorized access even if a password is compromised. While WPA3 improves resistance through mechanisms like Simultaneous Authentication of Equals (SAE), these enhancements are not sufficient on their own if weak passwords are used.

Finally, organizations should align their authentication practices with OWASP guidelines, particularly A07:2021 on Identification and Authentication Failures, to ensure that both protocol and user practices contribute to a robust security posture.

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